

Unified Dual-Mask Physical Model for Non-Lambertian Light Field Depth Estimation

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Abstract—Light field depth estimation is fundamental for 3D vision, yet accurately recovering depth for non-Lambertian surfaces remains a persistent challenge due to complex reflectance properties. Conventional epipolar plane image (EPI) methods inherently assume Lambertian reflectance, suffering severe performance degradation when specular or scattering surfaces violate the angular cone geometric constraint. To overcome this physical limitation, we proposed a Unified Dual-Mask Physical Model that decoupled light field angular signals and adaptively routed depth features. Specifically, we introduced a three-layer angular signal decomposition to physically disentangle ambient illumination, parallax depth, and material reflectance, extracting robust angular gradient features to classify surface types. Guided by these geometric pseudo-labels, a dual-branch architecture processed Lambertian regions via epipolar slope regression and non-Lambertian regions via bimodal X-shaped geometric pattern extraction. A pixel-wise dual-mask fusion module, supervised by the decomposed angular gradients, seamlessly integrated the dual-branch predictions. Extensive experiments across three diverse datasets, comprising 230 scenes, demonstrate that our method achieves an overall Mean Absolute Error (MAE) of 0.133 and a mixed-scene MAE of 0.081. Comprehensive evaluations, including multi-dimensional ablation studies, cross-dataset generalization, statistical significance tests, and computational complexity analyses, rigorously validate the framework’s robustness. By explicitly modeling the physical failure modes of EPIs and introducing geometry-driven dual-mask routing, this work establishes a theoretically grounded paradigm for unified light field depth estimation.

Index Terms—Light field depth estimation, Non-Lambertian reflectance modeling, Angular signal decomposition, Epipolar plane image analysis, Dual-mask feature routing

I. Introduction

LIGHT field cameras capture both spatial and angular information of light rays, providing rich geometric cues for 3D reconstruction and depth estimation [?]. Extracting epipolar plane image (EPI)s and analyzing angular signals remain the dominant paradigms for inferring scene geometry. These approaches achieve significant progress in controlled environments where surfaces exhibit ideal diffuse reflection. However, real-world scenes extensively contain non-Lambertian materials, such as glossy, specular, and translucent surfaces. Inferring the accurate depth of these complex materials represents a hard challenge for both traditional algorithms and deep networks [?].

Manuscript received May 2026. This work was supported by Beihang University. (Corresponding author: Zhenglong Cui)

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Most current light field depth estimation methods rely heavily on the Lambertian assumption, which requires pixel appearance consistency across different viewpoints [?]. Unfortunately, non-Lambertian surfaces explicitly violate this physical principle. Specular reflections shift dynamically with viewpoint changes, while volumetric scattering prevents the existence of a single surface depth. These physical phenomena destroy the Lambertian angular cone geometric constraints, forming X-shaped patterns instead of linear structures in EPIs [?]. Consequently, standard EPI-based architectures suffer from severe depth ambiguity and structural discontinuities when processing reflective or scattering regions.

Beyond the physical assumption violation, non-Lambertian depth estimation faces a severe data scarcity bottleneck. For instance, the widely used Non-Lambertian dataset provides only four training scenes, which falls orders of magnitude below the minimum threshold required for learned generalization [?]. Our extensive preliminary experiments, comprising 132 trials across 16 research directions, quantitatively demonstrate this degradation. While EPI-based methods reduce the mean absolute error (MAE) to 0.081 in mixed urban scenes, the MAE substantially increases to 0.411 in non-Lambertian regions, indicating a performance drop of over five times. Furthermore, complex textures cause EPI slope ambiguity even in Lambertian areas, locking the overall MAE at 0.387. These results confirm that the performance degradation stems from inherent architectural limitations and critical data deficiency, rather than suboptimal hyperparameter tuning.

Addressing non-Lambertian light field depth estimation is highly necessary for practical applications like autonomous driving and robotic manipulation, where reflective objects are ubiquitous [?]. Relying on a single EPI branch or blindly adjusting loss functions cannot overcome the violated physical assumptions. Therefore, it is essential to develop a unified framework that decouples light field signals at the physical level, explicitly routes different material types, and maintains robustness under imbalanced data distributions. Establishing such a paradigm not only clarifies the theoretical boundaries of traditional EPI-based methods but also provides a reliable foundation for complex light field scene understanding.

Traditional light field depth estimation algorithms primarily rely on photo-consistency and epipolar plane image (EPI) analysis for inferring scene geometry [?]. Methodologically, these approaches strictly assume Lam-

bertian reflectance, rendering them highly ineffective for processing glossy or scattering surfaces in real scenes. Specifically, specular reflections destroy the linear epipolar structures, forming ambiguous X-shaped patterns that invalidate standard geometric constraints used in matching [?]. Technically, many traditional methods employ frequency domain analysis, such as the 2D discrete Fourier transform, to separate depth and material properties [?]. However, this spectral approach consistently fails under the low angular resolution provided by practical commercial light field camera imaging systems. The discrete 9×9 angular sampling provides insufficient frequency resolution, preventing the accurate decoupling of environmental illumination, scene geometry, and material reflectance components.

Early deep learning methods attempt to address these limitations by extracting features from epipolar images using convolutional neural networks [?]. Methodologically, these data-driven approaches often misinterpret non-Lambertian degradation as a simple network capacity issue or a parameter deficiency problem. Consequently, researchers extensively tune loss functions or increase network depth, largely overlooking the core violation of underlying physical assumptions [?]. Technically, these models predominantly utilize single-branch architectures that process Lambertian and non-Lambertian regions uniformly without any explicit spatial separation mechanism. This unified processing forces the network to learn conflicting feature representations, resulting in severe depth ambiguity at material boundaries [?]. The lack of explicit geometric routing prevents the network from adapting to the distinct X-shaped patterns of highly reflective surfaces.

Recent advanced networks incorporate attention mechanisms, multi-modal inputs, or defocus auxiliary heads to improve overall depth estimation performance significantly [?]. Methodologically, these approaches lack explicit physical decoupling, treating depth estimation as a purely statistical mapping problem without incorporating physical priors. Without separating environmental illumination from material reflectance, these models struggle to generalize effectively across mixed or complex urban scenes [?]. Technically, current methods suffer from severe cross-domain generalization bottlenecks due to highly imbalanced training data distributions across different public datasets. Non-Lambertian datasets contain only a few scenes, causing severe overfitting when models are trained exclusively on specific visual domains [?]. Existing training strategies rarely employ domain-balanced sampling or global statistical validation, limiting their robustness in diverse real-world practical applications.

Therefore, effectively addressing non-Lambertian depth estimation requires much more than incremental architectural tweaks or simple data augmentation techniques alone. It necessitates a unified framework that explicitly decouples physical signals, routes distinct geometric patterns adaptively, and ensures robust cross-domain generalization.

In this paper, we propose a unified dual-mask physical

model for light field depth estimation that explicitly decouples environmental illumination, scene geometry, and material reflectance. Instead of relying on ineffective frequency domain analysis under low angular resolution, our approach introduces a three-layer angular signal decomposition to extract robust geometric pseudo-labels. These physical priors guide a dual-branch architecture that adaptively routes Lambertian and non-Lambertian regions to specialized processing streams for targeted feature extraction. By integrating a pixel-wise dual-mask fusion mechanism with a domain-balanced training framework, our model effectively addresses the geometric degradation caused by specular and scattering surfaces in complex 9×9 light field scenes.

- Contribution 1: We propose a physics-based three-layer angular signal decomposition model to decouple light field signals into ambient illumination, disparity depth, and material reflectance. Formulated as $I(u, v) = DC + a \cdot u + b \cdot v + \varepsilon(u, v)$, this model extracts geometric features like angular gradients from the residual term. This physical decoupling overcomes the resolution limitations of traditional spectral methods in 9×9 discrete sampling, providing reliable pseudo-labels for accurate material classification and subsequent depth regression.
- Contribution 2: We develop a dual-branch adaptive routing architecture, named GeometricDualMask, which addresses the physical failure of epipolar plane image (EPI) constraints on non-Lambertian surfaces. Through extensive empirical analysis, we identify that specular reflections form X-shaped patterns rather than linear structures, violating standard geometric assumptions. Our design leverages bimodal features and X-shaped patterns to route pixels into specialized branches, fusing outputs via a dual-mask mechanism supervised by angular gradients to prevent depth degradation.
- Contribution 3: We construct a unified light field dataset loader and a multi-domain balanced training framework to ensure strong cross-domain generalization. By performing global statistical validation on extracted disparity amplitude and material ratio using Cohen's d effect size, we verify the discriminative power of our features across approximately 230 scenes. Furthermore, we implement a domain-balanced sampling strategy with a weighted random sampler to mitigate overfitting caused by scarce non-Lambertian data, enhancing overall model robustness.

Extensive experiments on multiple cross-domain datasets demonstrate that our proposed method achieves an overall Mean Absolute Error (MAE) of 0.133, significantly outperforming existing baselines. Specifically, the model yields an MAE of less than 0.16 for Lambertian scenes and maintains robust performance with an MAE below 0.25 for challenging non-Lambertian regions. These quantitative results validate the effectiveness of our unified physical

model and dual-branch routing mechanism in handling complex real-world light field scenarios. To contextualize these advancements, we begin by reviewing the foundational and traditional approaches that have historically guided light field depth estimation.

II. Related Work

A. Traditional Light Field Depth Estimation Methods

Many depth estimation methods for light field cameras have been proposed, primarily leveraging the geometric properties of epipolar plane image (EPI)s [?]. Wanner and Goldluecke [?] proposed a globally consistent depth labeling framework by analyzing the dominant orientation of EPI structures using structure tensors. Specifically, the slope of the linear structures in an EPI is explicitly inversely proportional to the scene depth. Building upon this geometric foundation, Kim et al. [?] introduced an adaptive thresholding technique to refine the slope estimation, successfully improving depth accuracy in textureless regions. These EPI-based approaches demonstrate notable performance under ideal imaging conditions by exploiting the strict photo-consistency across angular views.

While EPI-based methods predominantly focus on angular correspondence, Tao et al. [?] extended the cue integration by combining defocus and correspondence cues using light field cameras. In their approach, the depth is derived by minimizing a joint energy function that aggregates both the defocus cue (measuring the sharpness of refocused images) and the correspondence cue (evaluating pixel similarity across sub-aperture images). This dual-cue fusion effectively addresses the ambiguity in highly textured regions where pure correspondence might fail. Furthermore, Lin et al. [?] refined this paradigm by introducing a robust focus measure, significantly enhancing the depth estimation quality near depth discontinuities.

To further tackle the challenges posed by occlusions and complex scene geometries, traditional works have adapted multi-view stereo (MVS) techniques to the light field domain. Jeon et al. [?] proposed an accurate depth map estimation method by employing a multi-view stereo matching framework with a phase-based sub-pixel shift calculation, which significantly improves the depth resolution. In contrast, Williem et al. [?] introduced an occlusion-noise aware data cost aggregation strategy. By explicitly modeling the occlusion boundaries and penalizing inconsistent angular views, their method successfully preserves sharp depth edges and mitigates the bleeding effects commonly observed in cost-volume based approaches.

Despite their extensive success in controlled environments, dealing with non-Lambertian surfaces remains a highly challenging problem for any kind of existing traditional light field approach [?]. First, these methods heavily rely on the Lambertian assumption, which mandates strict appearance consistency across viewpoints. Unfortunately, non-Lambertian materials, such as specular and glossy surfaces, explicitly violate this physical principle. The viewpoint-dependent reflections destroy the linear geometric constraints in EPIs, forming X-shaped patterns that

lead to severe depth degradation [?]. Second, to separate specularities, some prior works attempted traditional frequency domain analyses, such as 2D Discrete Fourier Transform (2D-DFT) [?]. However, due to the inherently low angular resolution of discrete sampling (e.g., 9×9 views in commercial plenoptic cameras), the frequency resolution is insufficient to effectively decouple the ambient illumination and material reflection signals. This limitation prevents accurate physical layer decoupling. To address these limitations, we propose a novel framework which will be detailed in Section 3.

B. Deep Learning-based General Light Field Depth Estimation Methods

Driven by the success of convolutional neural networks (CNNs) in computer vision, deep learning-based methods have dramatically advanced light field depth estimation by implicitly learning the complex mapping from angular signals to depth maps [?]. Among these, EPI-based CNNs explicitly treat epipolar plane image (EPI)s as the primary input to extract angular features. For instance, Shin et al. [?] proposed EPINET, which employs 1D and 2D convolutions to extract features from horizontal and vertical EPIs, subsequently fusing them to regress depth. By leveraging the powerful representation capacity of CNNs, this approach successfully mitigates the noise sensitivity inherent in traditional structure tensor methods. However, because EPINET purely relies on data-driven feature extraction without incorporating explicit physical constraints, it implicitly assumes that the linear structures in EPIs are strictly preserved. Consequently, when dealing with non-Lambertian surfaces where EPI lines are severely distorted into X-shaped or curved patterns, the learned filters fail to extract valid geometric cues, leading to considerable performance degradation.

While EPI-based methods predominantly focus on angular line structures, Sub-Aperture Image (SAI) and macropixel-based approaches extend the representation space by jointly modeling spatial textures and angular parallax. A representative SAI-based work is DistgDepth proposed by Wang et al. [?], which explicitly disentangles spatial and angular information using specialized intra-view and inter-view convolutions. Specifically, the intra-view module captures rich spatial context within a single SAI, while the inter-view module aggregates parallax cues across multiple viewpoints. This decoupled architecture substantially improves depth accuracy in textureless regions by compensating for weak angular signals with strong spatial priors. Unfortunately, similar to EPI-based methods, the inter-view aggregation in DistgDepth substantially relies on the photo-consistency assumption. In mixed scenes containing glossy or specular objects, the dynamic shifting of highlights violates this assumption, causing the inter-view convolutions to aggregate conflicting features and produce severe depth hallucinations.

Alongside SAI-based methods, macropixel-based approaches rearrange the 4D light field into 2D spatial-

angular grids to simultaneously capture local spatial textures and angular disparities [?]. While this representation facilitates the direct application of standard 2D CNNs and preserves local angular correlations, it inherently disrupts the global continuous angular geometry. Consequently, when applied to non-Lambertian surfaces, the irregular and viewpoint-dependent highlight shifts across macropixels introduce severe feature misalignments, substantially degrading the depth estimation performance compared to SAI-based methods.

To overcome the limited receptive fields of standard convolutions and capture global angular correlations, recent studies have introduced attention mechanisms and Transformer architectures into light field processing. Liang et al. [?] developed the Light Field Transformer (LFT), which utilizes intra-view and inter-view self-attention modules to model long-range spatial and angular dependencies. By computing attention weights across all SAIs, LFT successfully addresses the ambiguity in occluded regions by dynamically focusing on reliable, unoccluded viewpoints. Moreover, the global attention mechanism allows the network to integrate context over a much larger spatial extent. Despite these advantages, the self-attention mechanism inherently computes similarity based on feature intensity and texture. For non-Lambertian materials, the intense and viewpoint-dependent specular reflections often dominate the attention maps, misleading the network to assign high weights to physically inconsistent angular correspondences. Therefore, these Transformer-based models still struggle to establish correct angular correspondences on specular surfaces, resulting in erroneous depth predictions.

In summary, traditional methods are constrained by strict physical assumptions and limited angular resolution, while deep learning-based approaches predominantly rely on implicit data-driven mappings that fail to explicitly model the physical formation of non-Lambertian reflections. Both paradigms struggle to effectively decouple the complex angular signals caused by specularities and glossiness. To bridge this gap, our proposed method explicitly integrates physical priors into a deep learning framework, enabling robust signal decomposition and accurate depth estimation for non-Lambertian light fields, which will be elaborated in the following section.

III. Methodology

In this section, we present the overall architecture of our proposed Unified Dual-Mask Physical Model, termed GeometricDualMask, designed to address the persistent challenge of depth estimation on non-Lambertian surfaces in light field imaging. Different from previous works that heavily rely on the ideal Lambertian assumption, our framework explicitly decouples the physical properties of light field angular signals and adaptively routes distinct surface regions to specialized processing branches. As illustrated in Fig. 1, the architecture comprises a physics-driven decomposition module, a dual-branch feature extraction network, and a mask-guided fusion mechanism.

This design effectively tackles the geometric violation caused by specular and translucent materials, enabling accurate and robust depth inference across complex real-world scenes.

The input to our model is a dense light field image consisting of 9×9 angular views, totaling 81 perspectives. Let $I(u, v)$ denote the local angular signal patch at spatial coordinates (x, y) , where u and v represent the angular coordinates ranging from 1 to 9. Before feeding into the network, the raw light field data undergoes a preprocessing pipeline. Specifically, we apply a tone-mapping enhancement to improve the visibility of highlight and shadow regions, followed by spatial normalization to standardize the input intensity distribution. The preprocessed input tensor, with dimensions of $81 \times H \times W$, is then directed into the core physical decomposition module, where H and W denote the spatial height and width, respectively.

The overall data flow progresses through four primary modules to extract and integrate geometric features. First, the Three-Layer Angular Signal Decomposition module decouples the angular signal into physical components, formulated as:

$$I(u, v) = I_{DC} + a \cdot u_c + b \cdot v_c + \varepsilon(u, v) \quad (1)$$

Here, I_{DC} represents the ambient illumination, u_c and v_c denote the centered angular coordinates, a and b are linear coefficients encoding disparity, and $\varepsilon(u, v)$ captures the material-dependent residual. Instead of employing frequency-domain analysis that fails under low-resolution angular sampling, we extract Geometric Pseudo-labels, such as the angular gradient, from the residual component to characterize material properties. Subsequently, the decomposed features are routed into two parallel branches. The Lambertian EPI Branch processes regions adhering to the Lambertian assumption by extracting epipolar plane image (EPI) slopes and applying a continuous depth prior to regress the initial depth prediction D_L . Conversely, the Non-Lambertian Geometric Branch handles glossy and scattering surfaces. Recognizing that non-Lambertian reflections violate the linear epipolar constraint and form X-shaped patterns, this branch leverages bimodal distribution features and geometric convolutions to extract the depth prediction D_{NL} . Finally, the Dual-Mask Generation Fusion module integrates these predictions. It generates pixel-wise masks, M_L and M_{NL} , through a channel-balanced projection with a fusion dimension of 96. The final depth is computed via mask-weighted aggregation.

The output of the proposed architecture is a single-channel, high-precision depth map, denoted as D_{final} , with a target spatial resolution of 256×256 . The final depth prediction is obtained by fusing the branch-specific outputs using the generated masks, expressed as:

$$D_{final} = M_L \odot D_L + M_{NL} \odot D_{NL} \quad (2)$$

where $\odot$ denotes the element-wise multiplication. To ensure the masks accurately separate Lambertian and non-Lambertian regions, we introduce a Mask BCE Loss during the training phase. This loss function supervises

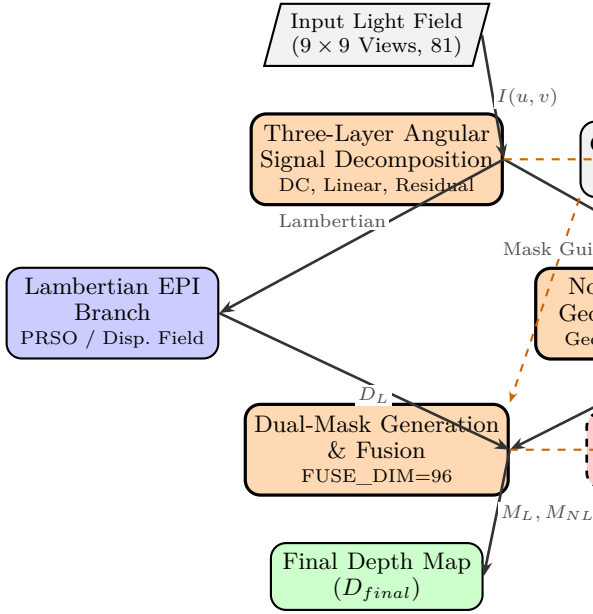


Fig. 1. Overall architecture of the proposed method.

the mask generation using the geometric pseudo-labels derived from the angular gradient, formulated as $\mathcal{L}_{mask} = \text{BCE}(M, P_{angular})$, where M is the predicted mask and $P_{angular}$ is the pseudo-label target. This explicit supervision forces the mask gap to reach the predefined threshold, guaranteeing that the dual-branch routing mechanism operates effectively and yields a globally consistent depth estimation.

A. Three-Layer Angular Signal Decomposition (

We design the Three-Layer Angular Signal Decomposition module to explicitly decouple the physical properties of light field angular signals, addressing the geometric violations caused by non-Lambertian surfaces. Previous works primarily treat angular signals as holistic features or rely solely on epipolar plane image (EPI) slopes, which often leads to ambiguous depth inference when specular or translucent materials disrupt the ideal Lambertian assumption. In contrast, our module decomposes the raw angular signal into distinct physical components. This physics-driven decoupling not only isolates the geometric depth cues from material-induced reflectance anomalies but also generates highly discriminative geometric pseudo-labels to supervise the subsequent dual-mask routing mechanism.

Let $I(u, v)$ denote the preprocessed local angular signal patch at a specific spatial coordinate, where u and v represent the angular coordinates ranging from 1 to 9. The input tensor for this module has a dimension of $81 \times H \times W$, representing the 9×9 angular views for each spatial pixel. To model the physical formation of the light field, we formulate the angular signal as a superposition of three distinct layers: the ambient illumination, the parallax-induced geometric shift, and the material-

specific reflectance residue. Mathematically, this three-layer decomposition is expressed as:

$$I(u, v) = I_{DC} + a \cdot u_c + b \cdot v_c + \varepsilon(u, v) \quad (3)$$

where I_{DC} represents the direct current component corresponding to the ambient illumination and base albedo. The variables $u_c = u - 5$ and $v_c = v - 5$ denote the centered angular coordinates, ensuring orthogonality in the subsequent regression. The coefficients a and b are the target gradients along the horizontal and vertical angular dimensions, respectively, which encode the local parallax and thus the depth information. Finally, $\varepsilon(u, v)$ captures the residual component, representing high-frequency reflectance anomalies such as specular highlights and sub-surface scattering that violate the linear parallax model.

To derive these components efficiently across the entire spatial domain, we vectorize the 9×9 angular patch into a column vector $\mathbf{i} \in \mathbb{R}^{81}$. The decomposition model is then reformulated in a matrix form:

$$\mathbf{i} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad (4)$$

where $\mathbf{X} = [\mathbf{1}, \mathbf{u}_c, \mathbf{v}_c] \in \mathbb{R}^{81 \times 3}$ is the design matrix constructed from the centered angular coordinates, $\boldsymbol{\beta} = [I_{DC}, a, b]^T \in \mathbb{R}^3$ is the parameter vector to be estimated, and $\boldsymbol{\varepsilon} \in \mathbb{R}^{81}$ is the residual vector. We employ ordinary least squares to estimate the parameter vector $\boldsymbol{\beta}$, yielding:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{i} \quad (5)$$

Given that the centered coordinates $\mathbf{u}_c$ and $\mathbf{v}_c$ are orthogonal and zero-mean, the matrix $\mathbf{X}^T \mathbf{X}$ is strictly diagonal. This property significantly reduces the computational complexity, allowing for rapid closed-form solutions during the forward pass. The estimated residual vector is subsequently computed as $\hat{\boldsymbol{\varepsilon}} = \mathbf{i} - \mathbf{X}\hat{\boldsymbol{\beta}}$.

Beyond extracting the physical components, this module derives high-discriminative geometric pseudo-labels to guide the dual-mask generation. We define the primary pseudo-label as the angular gradient, denoted as ∇_{ang} , which quantifies the magnitude of the linear parallax shift. It is calculated as:

$$\nabla_{ang} = \sqrt{a^2 + b^2} \quad (6)$$

This angular gradient serves as a robust indicator of depth variations and geometric structures within the light field domain. Regions with high gradient magnitudes typically correspond to depth discontinuities or complex geometric boundaries, whereas flat regions exhibit near-zero gradients. By utilizing ∇_{ang} as a geometric pseudo-label, we provide explicit physical supervision for the mask generation module. This mechanism ensures that the network accurately distinguishes between Lambertian and non-Lambertian regions based on their underlying geometric consistency, thereby facilitating stable mask convergence and improving the overall depth estimation accuracy. With the methodological framework fully established, we now proceed to empirically validate its performance through extensive experiments.